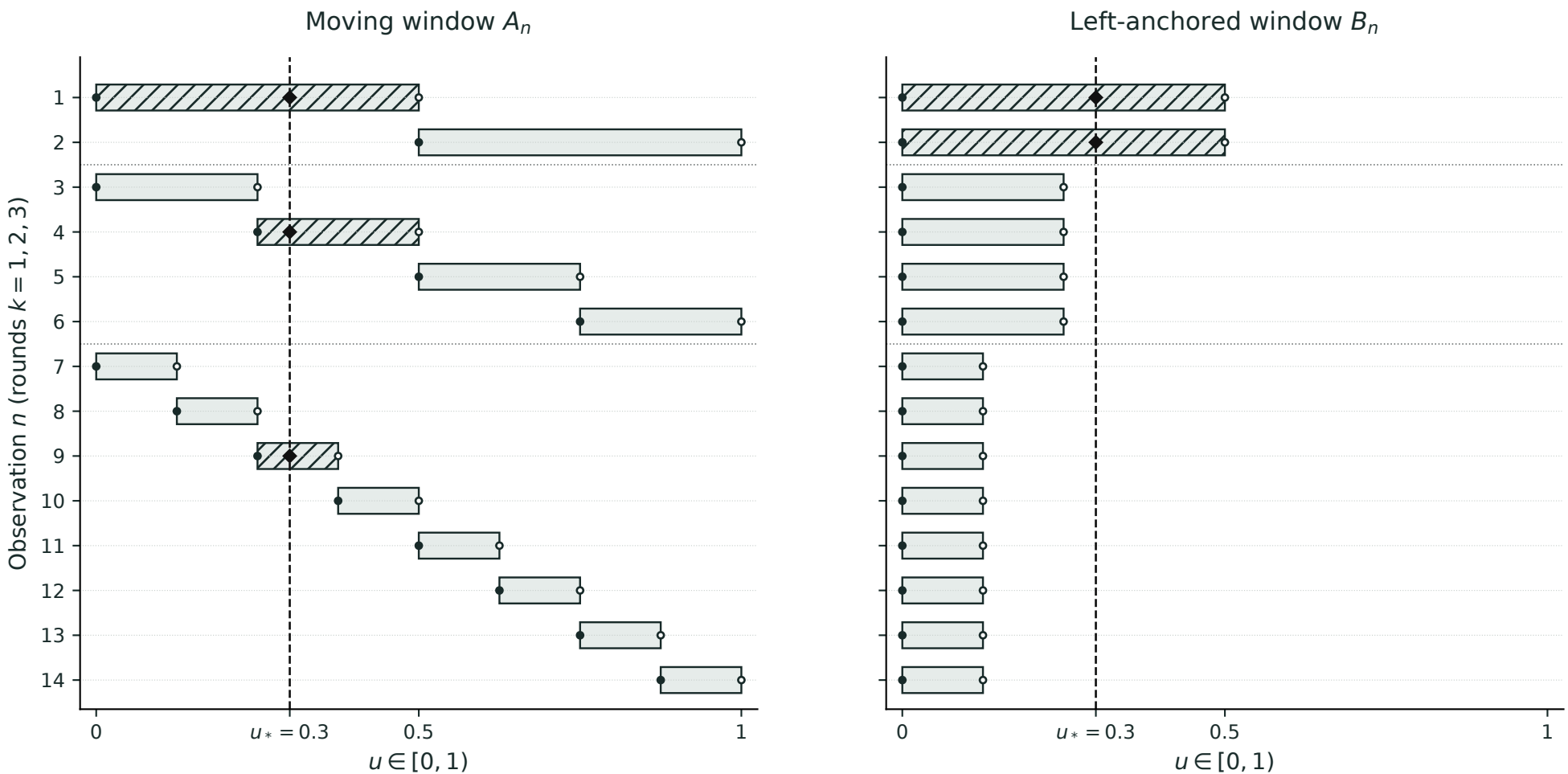

Filled/open circles mean included/excluded endpoints. Diamonds mark hits of $u_* = 0.3$.
Each moving round hits every u once; the fixed windows eventually miss every $u > 0$ ($u = 0$ is exceptional).